Name _____ Period _____ Date _____

6-6 Lesson Quiz: Exponential and Logarithmic Equations

short form; questions keep their numbers from the Savvas quiz Show your work.

2. The volume of one cloud, in cubic kilometers, grows according to the expression 4^{3x-1}, where x is time in hours. Another cloud grows according to the expression 8^x. After how many hours will the two clouds have the same volume? What is their volume, in cubic kilometers?

Ⓐ 0 hours; 1

Ⓑ $\frac{1}{2}$ hour; 2.83

Ⓒ $\frac{2}{3}$ hour; 4

Ⓓ 2 hours; 64

Work:

3. Max solved the equation $5^{x+3} = 25^{x-2}$, but made an error. His work is shown.

Part A Select each step that Max completed. Write the step next to the line it describes.

Line	Max's work	Step
1	$25^{x-2} = 5^{x+3}$	
2	$\left(5^2\right)^{x-2} = 5^{x+3}$	_____
3	$5^{2x-2} = 5^{x+3}$	_____
4	$2x - 2 = x + 3$	_____
5	$x = 5$	

Steps:

- Property of Equality for Exponential Equations
- Power of a Power Property
- Rewrite with a common base

Part B Identify and fix Max's mistake. Max made an error in line _____. The correct solution is $x =$ _____.

Corrected work: _____

4. Select **all** solutions of the equation $2\log x = \log(5x - 4)$.

☐ **A.** $x = 1$

☐ **B.** $x = 4$

☐ **C.** $x = 5$

☐ **D.** $x = 0$

☐ **E.** $x = -1$

Work (check each answer in the original equation):

Scoring: Question 3 (Parts A and B) and Question 4 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

6-6 Lesson Quiz, short form: answer key

Question numbers are the Savvas Lesson Quiz numbers (TE printed page 327A). Realize Skills Review and Practice codes are from the TE quiz item analysis; the TE prints no standards codes.

Q2. C, $\frac{2}{3}$ hour; 4. Common base 2: $4^{3x-1} = (2^2)^{3x-1} = 2^{6x-2}$ and $8^x = (2^3)^x = 2^{3x}$, so $6x - 2 = 3x$ and $x = \frac{2}{3}$. Volume: $8^{2/3} = 4$ (and $4^{3(2/3)-1} = 4^1 = 4$). Savvas item 2, DOK 2, skills F72, F73. Example 1. **Bonus 5:** 3 for C, 2 for the common-base work; a bare letter earns 3.

Q3. Part A: line 2 = Rewrite with a common base; line 3 = Power of a Power Property; line 4 = Property of Equality for Exponential Equations. **Part B: line 3;** $x = 7$. $\left(5^2\right)^{x-2} = 5^{2(x-2)} = 5^{2x-4}$, not 5^{2x-2}; then $2x - 4 = x + 3$, $x = 7$. Check: $5^{10} = 25^5$. Savvas item 3 (3A and 3B), DOK 3, skills F72, F73. Example 1; the packet's Practice is its structural parallel. **10 points:** Part A 4 (all three labels; 2 for two), Part B 6 (3 for line 3, 3 for $x = 7$ with work).

Q4. A and B ($x = 1$, $x = 4$). Power Property: $\log x^2 = \log(5x - 4)$, so $x^2 = 5x - 4$, $x^2 - 5x + 4 = 0$, $(x - 1)(x - 4) = 0$. Both make every argument positive ($x > 0$ and $5x - 4 > 0$): $x = 1$ gives $0 = 0$; $x = 4$ gives $\log 16 = \log 16$. C does not solve it; D and E make $\log x$ undefined. Savvas item 4, DOK 2, skills F72, F73. Example 5. **10 points:** 6 for exactly A and B (3 for one correct and none wrong), 4 for the work with the check.

Items left off this form (available for a retake or Form B): Savvas item 1, the solution to $7^{x-1} = 3$ (answer A, $x = \frac{\log 3}{\log 7} + 1$; DOK 1; skills F71, F72, F73; Example 3, cut in the fast packet); Savvas item 5, solve $\ln(3x) = 2x - 5$ by graphing and give the larger x to the nearest hundredth (answer 3.70; DOK 2; skills F72, F73; Example 6, cut). Neither item prints a graph, so nothing had to be redrawn.

IXL replacement: SmartScore 70 or above on YQY, BXU and RLX replaces this quiz with 20/20.